Samanvay Malapally Sudhakara

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Quantitative Associate with **2+ years** of experience applying advanced analytics, machine learning & AI to complex problems across finance, technology & business operations. Candidate for **CFA Level III** (Feb 2026) & the **FINRA SIE**. Adept at obtaining actionable insights from complex datasets using statistical analysis & predictive modeling, & at engaging & delivering to stakeholders. Proficient in cloud technologies & experienced in deploying scalable data pipelines & LLM-inference solutions. Strong quantitative & mathematical foundation built on graduate training in financial engineering, an undergraduate background in mechanical engineering & hands-on AI/ML expertise.

TECHNICAL SKILLS

Languages	Python, C++, Rust, SQL, R, Linux Shell
Data & Visualization	NumPy, pandas, Seaborn, Matplotlib, Plotly, ggplot2, Tableau, Power BI, Streamlit, Excel
Machine Learning	Regression, Classification, Tree-Based Ensemble Models, Reinforcement Learning (MAB, DQN, DDPG, PPO, SARSA)
Generative AI	LLM Training, Inference (vLLM, llama.cpp, SGLang, Ollama), Fine-Tuning (LoRA, QLoRA, GRPO), Quantization (GPTQ, AWQ), Speculative Decoding (MTP, DFlash), MCP Construction, Evals & Multi-Agent Orchestration
ML Packages	PyTorch, TensorFlow, JAX, scikit-learn, Transformers, Hugging Face Hub
Gen AI Packages	vLLM, MCP, Pydantic AI, LlamaIndex, Qdrant, Weaviate, RAG/GraphRAG, Phoenix/OpenInference, Claude Code, Codex
Cloud & Databases	GCP (Compute Engine, GCS, Vertex AI, Cloud Run, BigQuery, ADK), AWS (S3, EC2, Redshift, SageMaker, Lambda, Strands)
Data Engineering & CI/CD	Git, GitHub, GitLab, Docker, Snowflake, dbt, MySQL, Apache (Hadoop, Airflow, Spark, Kafka)
Quantitative Skills	Stochastic Calculus, Derivative Pricing, Hypothesis Testing, Statistical Analysis, Risk Management, Market Microstructure, Algorithmic Trading, Fixed Income & Interest Rate Models, Game Theory & Auction Theory

PROFESSIONAL EXPERIENCE

Quantitative Associate, *Northwestern Mutual Investment Management Company LLC, Milwaukee, WI* — Oct 2024 – Present

- Quant on the Public Fixed Income team ($140B+ AUM) spanning the full fixed income spectrum IG/HY corporates, EM sovereign & corporate credit, MBS, CMBS, CLOs, ABS, munis, rates & macro, and fixed income derivatives plus the affiliated CLO manager, 720 East (∼$4.2B AUM).
- Led the department's shift from Tableau, Power BI & Excel reporting to a Python/Streamlit stack for dashboards & tooling, enabling faster, more programmatic & interactive decision making.
- Built tooling that consolidates portfolio analytics from BlackRock Aladdin Explore, Bloomberg, EMBS & internal sources into unified Snowflake tables & views modeled with dbt, backed by a comprehensive data test suite & data integrity monitors.
- Architected a multi-app Streamlit interface mirroring Tableau's UI, fronted by a semantic layer, with apps that let PMs aggregate & slice the portfolio by any field & render it as a table, matrix, time series, scatter or treemap with fully customizable axes.
- Implemented on-the-fly bottom-up Brinson performance attribution & return stitching via the Frongello technique, computed over fully customizable run-time aggregations in seconds.
- Developed trader tools that parse Markit & Neptune quotes into an actionable format & link directly to the portfolio analytics, collapsing the ideation-to-trade workflow from hours to minutes.
- Built structured products tooling CPR & S-curves, WALA ramps over custom-built cohorts from loan-level data as part of the embedded quant program.
- Automated end-of-day report ingestion with Selenium & ChromeDriver, cutting daily data ingestion time by 4 hours.

Quantitative Analyst, *JIA Finance, New York, NY* — Jan 2023 – Aug 2023

- Automated mortgage guideline interpretation using the OpenAI GPT-3.5 & GPT-4 APIs with LangChain & FAISS/Chroma vector stores for rapid ingestion & validation of documents, improving the speed & accuracy of mortgage processing.
- Engineered specialized, task-specific Retrieval-Augmented Generation (RAG) Q&A chains that refined query processing through a recursive tree approach combined with LLM-based memoization, boosting zero-shot accuracy by 30%.
- Ran exploratory data analysis on over 100M Fannie Mae mortgage records in Tableau, with data cleaning & feature engineering via AWS SageMaker & the Redshift connector, to surface actionable insights.
- Modeled loan survival & default rate curves using the Cox proportional hazards model & the Kaplan–Meier estimator, & assessed the impact of macroeconomic factors such as unemployment & income on default & delinquency rates.
- Crafted a comprehensive cash flow model for mortgage-backed securities (MBS), integrating credit default, prepayment & loss severity models with a yield-curve-calibrated Cox–Ingersoll–Ross (CIR) interest rate model.

PROJECTS

High Frequency Trading Backtesting Pipeline with RL Trading Bot

- Built parsers for tick, depth-update & market-by-order data from sources such as CME Globex, IEX, Binance & OANDA, with cleaning & exploratory analysis to identify outliers, spot missing data & visualize the data (link)
- Simplified strategy backtesting with command-line pipelines driven by Linux shell scripts that integrate the C++ & Python components needed to run backtests, process results & tune hyperparameters (link)
- Engineered DQN & DDPG models in C++ for a market-taking agent & a market-making agent, using the PyTorch C++ API for the models & SQLite for the experience replay store, backtested on the pipelines above (link)

Forex Ladder Trading Strategy *(practicum sponsored by BP Trading)*

- Enhanced return potential by developing a forex ladder trading strategy using quantitative techniques for grid & lot-sizing optimization.
- Sped up backtests by up to 2000x using just-in-time compilation on tick data across major currency pairs.
- Tuned proprietary trading models with automated hyperparameter search (Optuna, PySwarms, DEAP genetic algorithms) & validated them using walk-forward analysis & Monte Carlo simulation.

Agentic Equity Research Pipeline on Self-Hosted LLM Inference

- Deployed a 35B-parameter MoE model (Qwen3.5-A3B) on an NVIDIA DGX Spark (GB10/ARM64, 128GB unified memory) via vLLM with NVFP4 quantization & DFlash speculative decoding, fronted by a Pydantic AI + MCP agent gateway (streamable HTTP) exposed through a Cloudflare Tunnel with Access JWT for zero-open-port ingress & instrumented with Phoenix/OpenInference.
- Engineered an OCR pipeline with built-in verification over a BSE/NSE filing sentinel, using PaddleOCR & LlamaParse (LlamaIndex) for document processing & the self-hosted Qwen model for agentic verification of extracted figures; cleansed data lands in BigQuery tables with a full observability & verification trace, exposed over REST API & MCP.
- Architected an agentic equity analyst with human user-bias conditioning that analyzes tickers using the self-hosted LLM, grounded on the cleansed financial data plus a Qdrant RAG engine & web search for the latest news, with Phoenix observability across the run, to generate research reports & recommendations.

CERTIFICATIONS

- **Bloomberg Market Concepts** — *Bloomberg for Education*
- **Machine Learning Specialization** — *DeepLearning.AI*
- **Deep Learning Specialization** — *DeepLearning.AI*

EDUCATION

MS Financial Engineering (GPA 3.77/4.00), *University of Illinois Urbana-Champaign, Champaign, IL* May 2024

BE Mechanical Engineering, *Ramaiah Institute of Technology, Bangalore, India* July 2021